Dickson College

Assessment Period:	**2026 S1**
Course:	**PHYSICS**
Unit:	**Unit 3: Gravity and Electromagnetism (1.0)**
Accreditation:	**T**
Year:	**12**

Specific Unit Goals

This unit should enable students to:

- understand that motion in gravitational, electric and magnetic fields can be explained using Newton's Laws of Motion
- understand how the electromagnetic wave model explains the production and propagation of electromagnetic waves across the electromagnetic spectrum
- understand transformations and transfer of energy in electromagnetic devices, as well as transformations and transfer of energy associated with motion in electric, magnetic and gravitational fields
- understand how models and theories have developed over time, and the ways in which physical science knowledge and associated technologies interact with social, economic, cultural and ethical considerations
- use science inquiry skills to design, conduct, analyse and evaluate investigations into uniform circular motion, projectile motion, satellite motion and gravitational and electromagnetic phenomena, and to communicate methods and findings
- use algebraic and graphical representations to calculate, analyse and predict measurable quantities related to motion, gravitational effects and electromagnetic phenomena
- evaluate, with reference to evidence, claims about motion, gravity and electromagnetic phenomena and associated technologies, and justify evaluations
- communicate physics understanding using qualitative and quantitative representations in appropriate modes and genres

Content Descriptions

Science Inquiry Skills

- identify, research and construct questions for investigation; propose hypotheses; and predict possible outcomes
- design investigations, including the procedure to be followed, the materials required, and the type and amount of primary and/or secondary data to be collected; conduct risk assessments; and consider research ethics
- conduct investigations, including the manipulation of force measurers and electromagnetic devices, safely, competently and methodically for the collection of valid and reliable data

- represent data in meaningful and useful ways, including using appropriate si units, symbols and significant figures; organise and analyse data to identify trends, patterns and relationships; identify sources of uncertainty and techniques to minimise these uncertainties; utilise uncertainty and percentage uncertainty to determine the uncertainty in the result of calculations, and evaluate the impact of measurement uncertainty on experimental results; and select, synthesise and use evidence to make and justify conclusions
- interpret a range of scientific and media texts, and evaluate processes, claims and conclusions by considering the accuracy and precision of available evidence; and use reasoning to construct scientific arguments
- select, construct and use appropriate representations, including text and graphic representations of empirical and theoretical relationships, vector diagrams, free body/force diagrams, field diagrams and circuit diagrams, to communicate conceptual understanding, solve problems and make predictions
- select, use and interpret appropriate mathematical representations, including linear and non-linear graphs and algebraic relationships representing physical systems, to solve problems and make predictions
- communicate to specific audiences and for specific purposes using appropriate language, nomenclature, genres and modes, including scientific reports

Science as a Human Endeavour

- ICT and other technologies have dramatically increased the size, accuracy and geographic and temporal scope of datasets with which scientists work
- models and theories are contested and refined or replaced when new evidence challenges them, or when a new model or theory has greater explanatory power
- the acceptance of science understanding can be influenced by the social, economic and cultural context in which it is considered
- people can use scientific knowledge to inform the monitoring, assessment and evaluation of risk
- science can be limited in its ability to provide definitive answers to public debate; there may be insufficient reliable data available, or interpretation of the data may be open to question
- international collaboration is often required when investing in large-scale science projects or addressing issues for the Asia-Pacific region
- scientific knowledge can be used to develop and evaluate projected economic, social and environmental impacts and to design action for sustainability

Science Understanding: Gravity and motion

- the movement of free-falling bodies in earth's gravitational field is predictable
- all objects with mass attract one another with a gravitational force; the magnitude of this force can be calculated using newton's law of universal gravitation
- objects with mass produce a gravitational field in the space that surrounds them; field theory attributes the gravitational force on an object to the presence of a gravitational field
- when a mass moves or is moved from one point to another in a gravitational field and its potential energy changes, work is done on or by the field
- gravitational field strength is defined as the net force per unit mass at a particular point in the field

- the vector nature of the gravitational force can be used to analyse motion on inclined planes by considering the components of the gravitational force (that is, weight) parallel and perpendicular to the plane
- projectile motion can be analysed quantitatively by treating the horizontal and vertical components of the motion independently
- when an object experiences a net force of constant magnitude perpendicular to its velocity, it will undergo uniform circular motion, including circular motion on a horizontal plane and around a banked track
- Newton's Law of Universal Gravitation is used to explain Kepler's laws of planetary motion and to describe the motion of planets and other satellites, modelled as uniform circular motion

Mathematical representations and relationships: Gravity and motion

- $w = mg$

  w = weight force, m = mass, g = acceleration due to gravity (gravitational field strength)

- $F = \frac{GMm}{r^2}$ and $g = \frac{F}{m} = \frac{GM}{r^2}$

  F = gravitational force, G = universal constant of gravitation $(6.67 \times 10^{-11} Nm^2 kg^{-2})$, M = Mass of first body, m = mass of second body, r = separation between the centres of mass of the two bodies, g = acceleration due to gravity

- $v_y = gt + u_y,\ y = \frac{1}{2}gt^2 + u_y t,\ v_y^2 = 2gy + u_y^2,\ v_x = u_x$ and $x = u_x t$

  y = vertical displacement, x = horizontal displacement, u_y = initial vertical velocity, v_y = vertical velocity at time t, u_x = initial horizontal velocity, v_x = horizontal velocity at time t, g = acceleration due to gravity, t = time into flight

- $v = \frac{2\pi r}{T}$

  v = tangential velocity, T = period

- $a_c = \frac{v^2}{r}$

  a_c = centripetal acceleration, v = tangential velocity, r = radius of the circle

- $F_{net} = \frac{mv^2}{r}$

  F_{net} = net force, m = mass of body undergoing uniform circular motion, v = tangential velocity, r = radius of the circle

- $\frac{T^2}{r^3} = \frac{4\pi^2}{GM}$

  T = period of satellite, M = mass of the central body, r = orbital radius, G = universal constant of gravitation $(6.67 \times 10^{-11} Nm^2 kg^{-2})$

Science Understanding: Electromagnetism

- electrostatically charged objects exert a force upon one another; the magnitude of this force can be calculated using Coulomb's Law
- point charges and charged objects produce an electric field in the space that surrounds them; field theory attributes the electrostatic force on a point charge or charged body to the presence of an electric field

- a positively charged body placed in an electric field will experience a force in the direction of the field; the strength of the electric field is defined as the force per unit charge
- when a charged body moves or is moved from one point to another in an electric field and its potential energy changes, work is done on or by the field
- current-carrying wires are surrounded by magnetic fields; these fields are utilised in solenoids and electromagnets
- the strength of the magnetic field produced by a current is called the magnetic flux density
- magnets, magnetic materials, moving charges and current-carrying wires experience a force in a magnetic field; this force is utilised in DC electric motors
- magnetic flux is defined in terms of magnetic flux density and area
- a changing magnetic flux induces a potential difference; this process of electromagnetic induction is used in step-up and step-down transformers, DC and AC generators, and AC induction motors
- conservation of energy, expressed as Lenz's law of electromagnetic induction, is used to determine the direction of induced current
- electromagnetic waves are transverse waves made up of mutually perpendicular, oscillating electric and magnetic fields
- oscillating charges produce electromagnetic waves of the same frequency as the oscillation; electromagnetic waves cause charges to oscillate at the frequency of the wave

Mathematical representations and relationships: Electromagnetism

- $F = \frac{1}{4\pi\epsilon_0}\frac{Qq}{r^2}$

  F = force, $\frac{1}{4\pi\epsilon_0}$ = Coulomb constant ($9 \times 10^9 Nm^2C^{-2}$), q = charge on the first object, Q = charge on the second object, r = separation between the charges

- $E = \frac{F}{q} = \frac{1}{a\pi\epsilon_0}\frac{q}{r^2}$

  E = electric field strength, F = force, q = charge, r = distance from the charge, $\frac{1}{4\pi\epsilon_0}$ = Coulomb constant ($9 \times 10^9 Nm^2C^{-2}$)

- $V = \frac{\Delta U}{q}$

  V = electrical potential difference, ΔU = change in potential energy, q = charge

- $B = \frac{\mu_o I}{2\pi r}$

  B = magnetic flux density, I = current in wire, r = distance from the centre of the wire, $\frac{\mu_0}{2\pi}$ = magnetic constant $2 \times 10^{-7} TA^{-1}m$

- For a straight, current carrying wire perpendicular to a magnetic field $F = BIL$

  B = magnetic flux density, F = force on the wire, L = length of wire in the magnetic field, I = current in wire

- For a charge moving perpendicular to a magnetic field, $F = qvB$

  F = force on a charge moving in an applied magnetic field, q = charge, v =velocity of the charge, B = magnetic flux density

- $\Phi = BA\perp$

  Φ = magnetic flux, $A\perp$ = area of current loop perpendicular to the applied magnetic field, B = magnetic flux density

- $emf = -\frac{n\Delta(BA\perp)}{\Delta t} = -n\frac{\Delta\Phi}{\Delta t}$

  emf = induced potential difference, $\Delta\Phi$ = change in magnetic flux, *n* = number of windings in the loop, $A\perp$ = area of current loop perpendicular to the applied magnetic field, Δt = time interval over which the magnetic flux change occurs, B = magnetic flux density

- $\frac{V_p}{V_s} = \frac{n_p}{n_s}$

  V_p = potential difference across the primary coil, V_s = potential difference across the secondary coil, n_p = number of turns on primary coil, n_s = number of turns on secondary coil

- $I_pV_p = I_sV_s$

  I_p = current in the primary coil, V_p = potential difference across the primary coil, I_s = current in the secondary coil, V_s = potential difference across the secondary coil

Assessment Tasks

Name	Due Date	Weighting
Laboratory Task	10 March - 13 March	40%
Test	20 March - 24 March	30%
Test	15 June - 17 June	30%

School Assessment Information

Penalties for late and non-submission of work

See BSSS Policy and Procedure Manual 4.2.9.1 for further information.

Academic integrity

See BSSS Policy and Procedure Manual 4.2.10 for further information.

Appeals processes

See BSSS Policy and Procedure Manual 7.3 for further information.

Moderation procedures (internal and external)

See BSSS Policy and Procedure Manual 5 for further information.

Meshing procedures

See BSSS Policy and Procedure Manual 5.4.1 for further information.

Method of unit score calculation

See BSSS Policy and Procedure Manual 4.2.6.2 for further information.

Procedures for calculating course scores

See BSSS Policy and Procedure Manual 4.2.11.2 for further information.

Achievement Standards for PHYSICS T - Year 12

	*A student who achieves an **A** grade typically*	*A student who achieves a **B** grade typically*	*A student who achieves a **C** grade typically*	*A student who achieves a **D** grade typically*	*A student who achieves an **E** grade typically*
Concepts, Models & Applications	● evaluates the properties and functions of system components, processes and interactions, and the interplay and effects of factors across a range of scales ● evaluates applications, limitations, and predictions of theories and models to explain systems and create solutions, with evidence, in unfamiliar contexts ● evaluates evidence with reference to analysis of models and/or theories, and develops evidence-based conclusions and evaluates limitations	● analyses the properties and functions of system components, processes and interactions, and the interplay and effects of factors across a range of scales ● analyses applications, limitations, and predictions of theories and models to explain systems and create plausible solutions, with evidence in familiar contexts ● analyses evidence with reference to models and/or theories, and develops evidence-based conclusions and discusses limitations	● explains the fundamental properties and functions of system components, processes and interactions, and the effects of factors across a range of scales ● explains applications, limitations, and predictions of theories and models to explain systems and create plausible solutions in familiar contexts ● describes evidence with reference to models and/or theories, and develops evidence-based conclusions and identifies limitations	● describes the fundamental properties and functions of system components, processes and interactions, and the effects of one or more factors ● describes the nature, functions, limitations and applications of theories and models to create solutions to problems with supporting evidence ● describes evidence, and develops conclusions with some reference to models and/or theories	● identifies the fundamental properties and functions of system components, processes and interactions, and some affective factors ● identifies the nature, functions, limitations and applications of theories and models, and suggest solutions to problems with supporting evidence ● identifies evidence, and asserts conclusions with little or no reference to models and/or theories
Contexts	● evaluates epistemology, role of peer review, collaboration, and technology in developing knowledge ● evaluates the influence of social, economic, ethical and cultural factors on Science	● analyses epistemology, role of peer review and technology in developing knowledge ● analyses the influence of social, economic, ethical and cultural factors on Science	● explains epistemology, role of peer review and technology in developing knowledge ● explains the influence of social, economic, ethical and cultural factors on Science	● describes role of peer review and technology in developing knowledge ● describes the influence of social, economic, ethical and cultural factors on Science	● identifies that scientific knowledge has changed over time ● identifies the influence of social, economic, ethical and cultural factors on Science
Inquiry Skills	● designs, conducts and improves safe, ethical and original inquiries individually and collaboratively, that collect valid, reliable data in response to a complex question ● evaluates cause and correlation, anomalies, reliability and validity of data and representations, and critically analyses errors ● evaluates processes and claims, and provides a critique based on evidence, and evaluates alternatives ● reflects with insight on own thinking and that of others, evaluates planning, time management, and use of appropriate independent and collaborative work strategies ● communicates concisely, effectively and accurately, with scientific literacy in a range of modes, representations, and genres for specific audiences and purposes, and accurate referencing	● designs, conducts and improves safe, ethical inquiries individually and collaboratively, that collect valid, reliable data in response to a question ● analyses cause and correlation, anomalies, reliability and validity of data and representations, and analyses errors ● explains processes and claims, and provides a critique with reference to evidence, and analyses alternatives ● reflects on their own thinking and analyses planning, time management, and use of appropriate independent and collaborative work strategies ● communicates clearly and accurately, with scientific literacy in a range of modes, representations and genres for specific audiences and purposes, and accurate referencing	● plans and conducts safe, ethical inquiries individually and collaboratively, that collect valid data in response to a familiar question ● describes causal and correlational relationships, anomalies, reliability and validity of data and representations, and discusses common errors ● describes processes and claims, and identifies alternatives with reference to reliable evidence ● reflects on their own thinking and explains planning, time management, and use of appropriate independent and collaborative work strategies ● communicates accurately demonstrating scientific literacy, in a range of modes, representations, and genres for specific purposes, and mostly consistent referencing	● follows a procedure to conduct safe, ethical inquiries individually and collaboratively, to collect data in response to a simple question with varying success ● describes trends, relationships and anomalies in data, identifies anomalies, and cites sources of error ● describes processes and claims, and identifies the need for improvements with some reference to evidence ● reflects on their own thinking, with reference to planning and the use of appropriate independent and collaborative work strategies ● communicates demonstrating some scientific literacy, in a range of modes, representations, and genres with some evidence and inconsistent referencing	● follows a procedure to conduct safe, ethical inquiries individually and collaboratively, to collect data with little or no connection to a question ● identifies trends and relationships in data with reference to sources of error ● identifies processes and the need for some improvements, with little or no reference to evidence ● reflects on their own thinking with little or no reference to planning, time management, and use of appropriate independent and collaborative work strategies ● communicates demonstrating limited scientific literacy, in a range of modes and representations, with inconsistent and inaccurate referencing